## VALVE, VALVE SPRING, VALVE SEAL, VALVE GUIDE [SKYACTIV-D 1.5]

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### Purpose, Function

**Valve**

- The intake valve is pressed down by the cam, the intake air passage is opened, and air is introduced into the cylinder.
- The exhaust valve is pressed down by the cam, the exhaust air passage is opened, and combustion gas is exhausted from the cylinder.

**Valve spring**

- The valve spring adheres the valve to the valve seat by pulling up the valve by spring force, and air tightness in the combustion chamber is maintained.

**Valve seal**

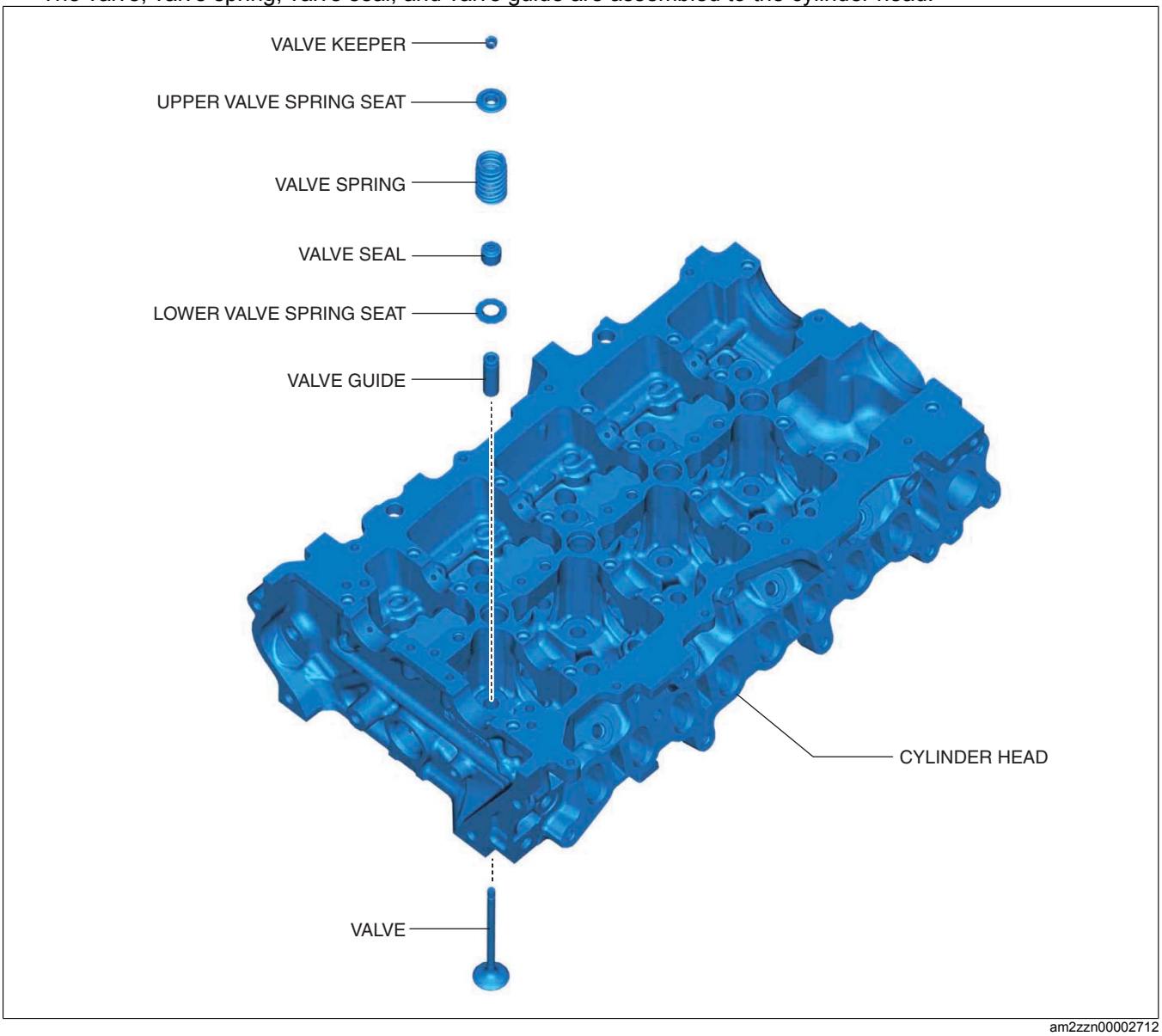
- The valve seal prevents engine oil from entering combustion chamber through the valve guide, and engine oil consumption is suppressed.

**Valve guide**

- The valve guide maintains the valve stem to open and close the valve smoothly.
- The valve guide cools the valve by transmitting valve heat to the cylinder head.

### Construction

- The valve, valve spring, valve seal, and valve guide are assembled to the cylinder head.




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- The sliding resistance has been reduced by setting the valve spring load to the required minimum.

- The intake valve and exhaust valve are subjected to a nitriding treatment to enhance wear resistance.
- Unequal pitch valve spring has been adopted.
- The valve guide is made of sintered alloy with excellent wear resistance.